

Evaluation in Polar Co-ordinates

$$x = r \cos \theta$$

$$y = r \sin \theta$$

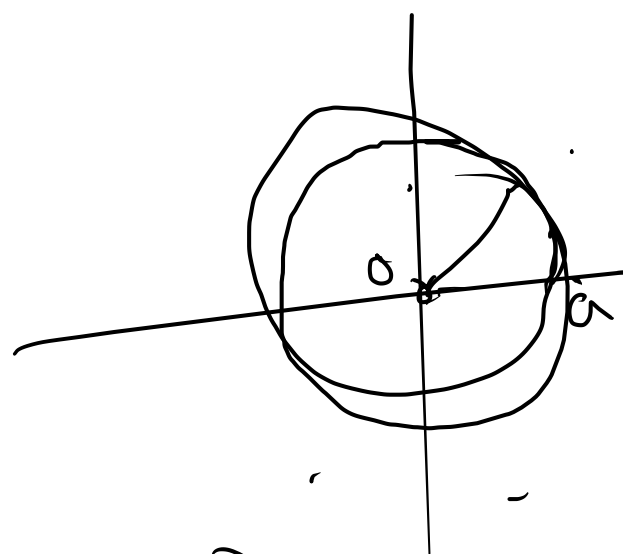
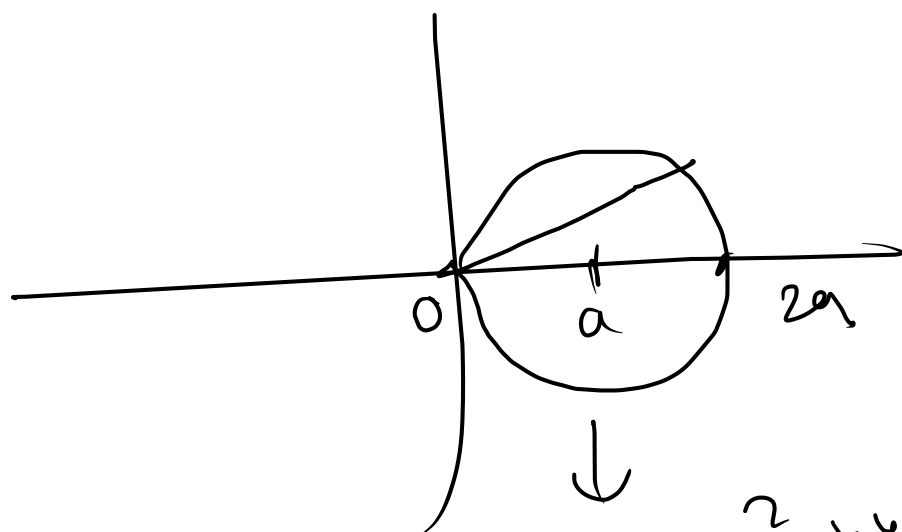
$$r^2 = x^2 + y^2$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right)$$

$$dx dy = r dr d\theta$$



$$\int \int f(x, y) dy dx \rightarrow \int_{\theta=\theta_1}^{\theta_2} \int_{r=f_1(\theta)}^{r=f_2(\theta)} f(r \cos \theta, r \sin \theta) r dr d\theta$$



$$\underline{\underline{x^2 + y^2 = a^2}}$$

$$r^2 = a^2$$

$$\underline{\underline{x^2 + y^2 - 2x = 0}}$$

$$r^2 = 2r \cos \theta$$

1. $\iint r^3 dr d\theta$ over the area included between the circles $r = 2 \sin \theta$ and $r = 4 \sin \theta$

$$x = r \cos \theta, \quad y = r \sin \theta$$

$$x^2 + y^2 = r^2$$

$$r = 2 \sin \theta$$

$$r^2 = 2r \sin \theta$$

$$x^2 + y^2 = 2y$$

$$x^2 + y^2 - 2y + \frac{4}{4} = 1 \Rightarrow x^2 + (y-1)^2 = 1$$

Center $(0, 1)$ Rad 1

$$r = 4 \sin \theta$$

$$r^2 = 4r \sin \theta \Rightarrow x^2 + y^2 = 4y \Rightarrow x^2 + y^2 - 4y + 4 = 4$$

$$= x^2 + (y-2)^2 = 2^2$$

Center $(0, 2)$ Rad 2

$$\int_0^\pi \int_{2 \sin \theta}^{4 \sin \theta} r^3 dr d\theta$$

$$= \int_0^\pi \left(\frac{r^4}{4} \right)_{2 \sin \theta}^{4 \sin \theta} d\theta$$

$$= \int_0^\pi \left(\frac{4^4 \sin^4 \theta}{4} - \frac{2^4 \sin^4 \theta}{4} \right) d\theta$$

$$= \int_0^\pi (64 \sin^4 \theta - 16 \sin^4 \theta) d\theta$$

$$= \int_0^\pi 48 \sin^4 \theta d\theta = 48 \int_0^\pi \sin^4 \theta d\theta$$

$$f(\theta) = \sin^4 \theta$$

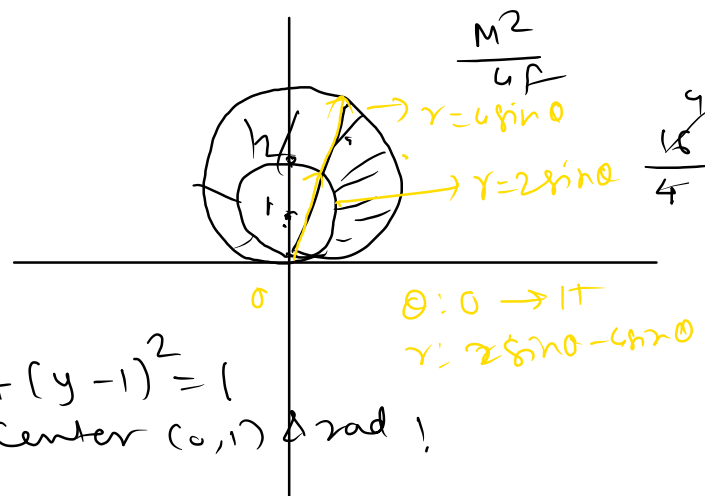
$$f(\pi - \theta) = \sin^4(\pi - \theta) = \sin^4 \theta = f(\theta)$$

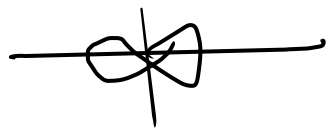
$$= 2 \times 48 \int_0^{\pi/2} \sin^4 \theta d\theta$$

$$= 2 \times 48 \times \frac{1}{2} B\left(\frac{5}{2}, \frac{1}{2}\right)$$

$$= 60 \frac{\Gamma(5/2) \Gamma(1/2)}{\Gamma(3)} = \frac{60}{2} \times \frac{3}{2} \times \frac{1}{2} \left(\frac{\sqrt{\pi}}{2}\right)^2$$

$$= \frac{45\pi}{2}$$





2. Evaluate $\iint \frac{r dr d\theta}{\sqrt{r^2 + a^2}}$ over one loop of the lemniscate $r^2 = a^2 \cos 2\theta$

$$\theta: -\pi/4 \rightarrow \pi/4$$

$$r: 0 \rightarrow a\sqrt{\cos 2\theta}$$

$$\int_{-\pi/4}^{\pi/4} \int_0^{a\sqrt{\cos 2\theta}} \frac{r dr d\theta}{\sqrt{r^2 + a^2}}$$

$$r^2 + a^2 = t$$

$$2r dr = dt$$

$$r dr = dt/2$$

$$r: 0 \rightarrow a\sqrt{\cos 2\theta}$$

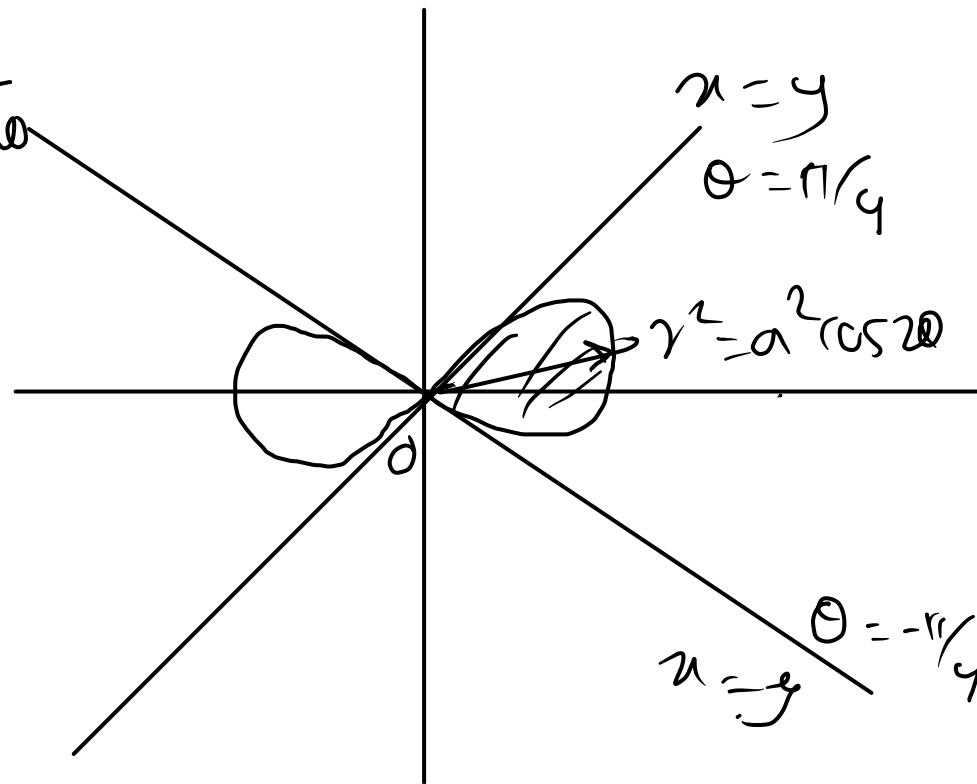
$$t: a^2 \rightarrow a^2 \cos 2\theta + a^2 = a^2 (\cos 2\theta + 1) = a^2 \cos^2 \theta$$

$$\int_{-\pi/4}^{\pi/4} \int_{a^2}^{a^2 \cos^2 \theta} \frac{dt}{2\sqrt{t}} d\theta$$

$$\frac{dt}{2\sqrt{t}} d\theta$$

$$= \left(\frac{2 - \pi/2}{2} \right) a$$

$$\left(2 - \frac{\pi}{2} \right) a$$



3. Evaluate $\iint r e^{-r^2/a^2} \cos \theta \sin \theta \, d\theta dr$ over the upper half of the circle $r = 2a \cos \theta$

$$x = r \cos \theta, \quad y = r \sin \theta$$

$$r = 2a \cos \theta$$

$$r^2 = 2ar \cos \theta$$

$$x^2 + y^2 = 2ax$$

$$x^2 - 2ax + a^2 + y^2 = a^2$$

$$(x-a)^2 + y^2 = a^2$$

Center $(a, 0)$ & radius a

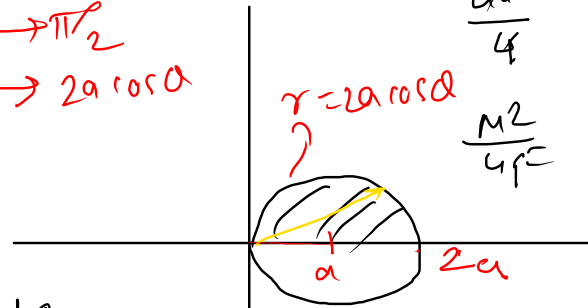
$$\theta: 0 \rightarrow \pi/2$$

$$r: 0 \rightarrow 2a \cos \theta$$

$$r = 2a \cos \theta$$

$$\frac{4a^2}{4}$$

$$\frac{M^2}{4f^2}$$



$$\int_0^{\pi/2} \int_0^{2a \cos \theta} r e^{-r^2/a^2} \cos \theta \sin \theta \, dr d\theta$$

$$r^2/a^2 = t \Rightarrow 2r dr = a^2 dt$$

$$r: 0 \rightarrow 2a \cos \theta$$

$$t: 0 \rightarrow 4 \cos^2 \theta$$

$$= \int_0^{\pi/2} \cos \theta \sin \theta \int_0^{4 \cos^2 \theta} e^{-t} \frac{a^2}{2} dt d\theta$$

$$= \frac{a^2}{2} \int_0^{\pi/2} \cos \theta \sin \theta \left(\frac{e^{-t}}{-1} \right)_0^{4 \cos^2 \theta} d\theta$$

$$= \frac{a^2}{2} \int_0^{\pi/2} \cos \theta \sin \theta (-e^{-4 \cos^2 \theta} + 1) d\theta$$

$$\text{put } 4 \cos^2 \theta = t$$

$$-8 \cos \theta \sin \theta d\theta = dt$$

$$\cos \theta \sin \theta d\theta = -\frac{dt}{8}$$

$$\theta: 0 \rightarrow \pi/2$$

$$t: 4 \rightarrow 0$$

$$= \frac{a^2}{2} \int_4^0 (1 - e^{-t}) \left(-\frac{dt}{8} \right) = \frac{a^2}{16} \int_0^4 (1 - e^{-t}) dt$$

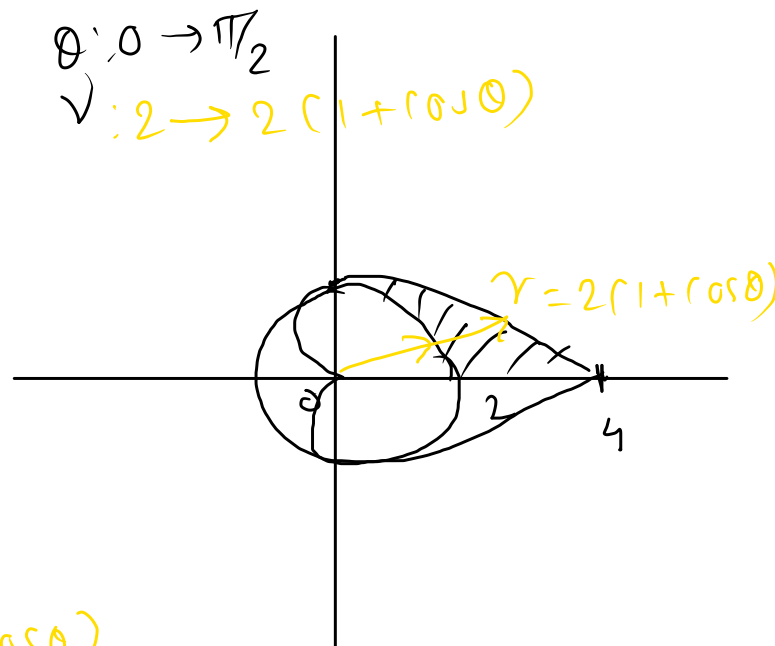
$$= \frac{a^2}{16} \left[t + e^{-t} \right]_0^4$$

$$= \frac{a^2}{16} [4 + e^{-4} - 1]$$

4. Evaluate $\iint_R \sin \theta \, dA$ where R is the region in the first quadrant that is outside the circle $r = 2$ and inside the cardioid $r = 2(1 + \cos \theta)$

$$\begin{aligned} x &= r \cos \theta \\ y &= r \sin \theta \\ dxdy &= r dr d\theta \\ r &= 2 \\ r^2 &= 2^2 \\ x^2 + y^2 &= 2^2 \end{aligned}$$

$$\begin{aligned} &\iint_R \sin \theta \, dA \\ &= \int_0^{\pi/2} \int_2^{2(1+\cos \theta)} \sin \theta \, r dr d\theta \\ &= \int_0^{\pi/2} \sin \theta \left(\frac{r^2}{2} \right)_2^{2(1+\cos \theta)} d\theta \end{aligned}$$



$$= \int_0^{\pi/2} \sin \theta \left(\frac{4(1+\cos \theta)^2}{2} - \frac{4}{2} \right) d\theta$$

$$= 2 \int_0^{\pi/2} \sin \theta (1 + 2\cos \theta + \cos^2 \theta - 1) d\theta$$

$$= 2 \int_0^{\pi/2} (2\sin \theta \cos \theta + \sin \theta \cos^2 \theta) d\theta$$

$$= 2 \left[2 \cdot \frac{1}{2} \beta(1,1) + \frac{1}{2} \beta(1, 3/2) \right]$$

$$\begin{aligned} &= 2 \left[\frac{\pi \cdot \pi}{\pi^2} + \frac{1}{2} \frac{\pi \sqrt{3/2}}{\sqrt{5/2}} \right] = 2 \left[1 + \frac{1}{2} \frac{\sqrt{3/2}}{\sqrt{5/2}} \right] \\ &= 2 \left(1 + \frac{1}{3} \right) = \frac{8}{3} \end{aligned}$$

